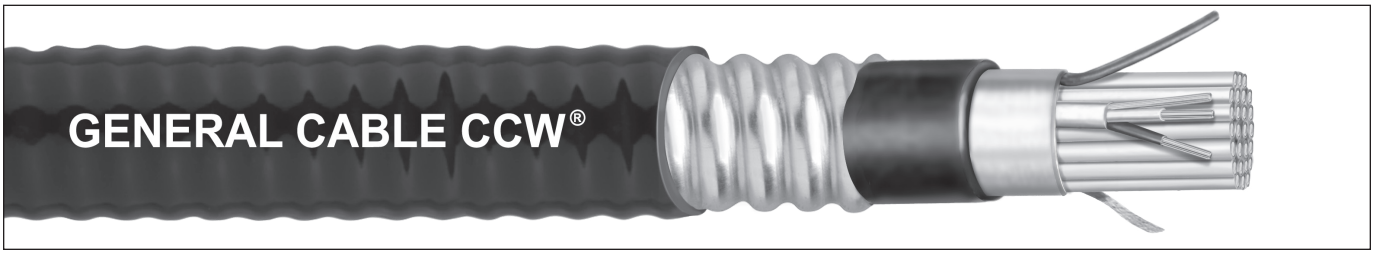
CCW® Armored Instrumentation, Pairs/Triads, Individual and Overall Shield

UL Type ITC-HL/PLTC, PVC, 300 V, 105°C, Sunlight-Resistant, Direct Burial
UL Marine Shipboard Cable, ABS CWCMC



Product Construction:

Conductor:
- Bare annealed copper per ASTM B3
- Class B stranding per ASTM B8

Insulation:
- Flame-retardant Polyvinyl Chloride (PVC), rated 105°C per UL Standards 13 and 2250
- Color-coded per ICEA Method 1: pairs – black and white; triads – black, white and red. Each conductor in each pair or triad is printed alphanumerically for easy identification

Shielded Pairs/Triads:
- Isolated and individually twisted pairs or triads with a Flexfoil® aluminum/polyester tape shield providing 100% coverage
- Stranded tinned copper drain wire, two sizes smaller than insulated conductors

Cable Assembly:
- Individually shielded pairs or triads and communication wire are cabled together with a left hand lay
- Communication wire: 22 AWG solid bare copper, flame-retardant Polyvinyl Chloride (PVC), rated 105°C, orange

Overall Shield:
- Flexfoil® aluminum/polyester tape shield providing 100% coverage
- Stranded tinned copper drain wire, same size as insulated conductors

Inner Jacket:
- Flame-retardant Polyvinyl Chloride (PVC) per UL Standards 13 and 2250, black
- Low temperature performance meets ASTM D746 brittleness temperature at or below -40°C
- Nylon rip cord to facilitate jacket removal

CCW Armor:
- Impervious, continuously welded and corrugated aluminum alloy sheath per UL 1569
- CCW armor conductivity meets the grounding requirements of NEC Article 250

Overall Jacket:
- Flame-retardant, moisture- and sunlight-resistant Polyvinyl Chloride (PVC) per UL Standards 13 and 2250, black
- Low temperature performance meets ASTM D746 brittleness temperature at or below -40°C

Applications:

- CCW armored Instrumentation cables with individually shielded pairs or triads and an overall shield provide superior protection and reliability against physical damage for use in instrumentation and process control applications requiring ITC-HL or PLTC wiring methods where shielding against both external EMI and EMI between groups is required
- For use as Power Limited Tray Cable on circuits rated 150 V or less and 5 amps or less in Class 2 or Class 3 circuits in accordance with NEC Article 725
- For use as Instrumentation Tray Cable on circuits rated 150 V or less and 5 amps or less in accordance with NEC Article 727
- Recognized for use in Class I, II, and III, Divisions 1 and 2; or Class I, Zones 1 and 2 hazardous locations per NEC Articles 501, 502, 503, and 505

Applications: (cont'd)

- Installed indoors or outdoors, in wet or dry locations, in a raceway, as aerial cable on a messenger, in cable trays, or for direct burial
- Recognized for use on fixed or floating offshore petroleum facilities as recommended by the American Petroleum Institute

Features:

- CCW armor provides superior mechanical protection and an impervious barrier to moisture, gas and liquids
- CCW armor provides EMI shielding performance
- Meets cold impact at -40°C

Specifications:

Design Adherence:
- UL 13 Power-Limited Circuit Cables
- UL 2250 Instrumentation Tray Cable
- UL 1569 Metal Clad Cables
- UL 1309/CSA C22.2 No. 245 Marine Shipboard Cable

Flame Tests:
- ICEA T-29-520 (210,000 BTU/hr)
- IEEE 383 (70,000 BTU/hr)
- CSA FT4
- IEEE 1202 (70,000 BTU/hr)
- UL 1581 (70,000 BTU/hr)
- IEC 60332-3 Cat. A

Compliances:
- UL Type PLTC, SUN RES, DIR BUR, -40°C, UL File # E36118
- UL Type ITC-HL, UL File # E177408
- UL Listed Marine Shipboard, UL File # E85994
- American Bureau of Shipping (ABS) Listed for CWCMC

General Cable
Phone: 888-593-3355
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UL Type ITC-HL/PLTC, PVC, 300 V, 105°C, Sunlight-Resistant, Direct Burial
UL Marine Shipboard Cable, ABS CWCMC

PART NUMBER	COND. SIZE (AWG)	NO. OF PAIRS	INSULATION THICKNESS		COMMUNICATION WIRE			INNER JACKET THICKNESS		NOMINAL CORE O.D.		NOMINAL ARMOR O.D.		JACKET THICKNESS		NOMINAL OVERALL O.D.		CROSS-SECTIONAL AREA[1]	APPROXIMATE NET WEIGHT	
					SIZE	INS. THICKNESS														
			mils	mm	AWG	mils	mm	mils	mm	INCHES	mm	INCHES	mm	mils	mm	INCHES	mm	(SQ. IN.)	LBS/1000 FT	kg/1000 m
20 AWG 7W (0.52 mm²) INDIVIDUAL AND OVERALL SHIELDED PAIRS																				
9250.20021221	20	2	15	0.38	22	12	0.30	40	1.02	0.36	9.1	0.58	14.7	50	1.27	0.69	17.5	0.38	202	301
9250.20041221	20	4	15	0.38	22	12	0.30	50	1.27	0.43	10.9	0.62	15.7	50	1.27	0.73	18.5	0.42	239	355
9250.20061221	20	6	15	0.38	22	12	0.30	50	1.27	0.48	12.2	0.71	18.0	50	1.27	0.82	20.8	0.54	292	434
9250.20081221	20	8	15	0.38	22	12	0.30	50	1.27	0.53	13.5	0.75	19.1	50	1.27	0.86	21.8	0.59	323	481
9250.20101221	20	10	15	0.38	22	12	0.30	50	1.27	0.57	14.5	0.80	20.3	50	1.27	0.91	23.1	0.66	401	597
9250.20121221	20	12	15	0.38	22	12	0.30	60	1.52	0.63	16.0	0.84	21.3	50	1.27	0.95	24.1	0.72	439	653
9250.20161221	20	16	15	0.38	22	12	0.30	60	1.52	0.72	18.3	0.97	24.6	50	1.27	1.08	27.4	0.93	511	760
9250.20201221	20	20	15	0.38	22	12	0.30	60	1.52	0.81	20.6	1.06	26.9	50	1.27	1.17	29.7	1.09	593	882
9250.20241221	20	24	15	0.38	22	12	0.30	70	1.78	0.90	22.9	1.15	29.2	50	1.27	1.26	32.0	1.26	718	1,069
9250.20361221	20	36	15	0.38	22	12	0.30	70	1.78	1.04	26.4	1.34	34.0	50	1.27	1.45	36.8	1.67	925	1,377
9250.20501221	20	50	15	0.38	22	12	0.30	70	1.78	1.19	30.2	1.51	38.4	60	1.52	1.65	41.9	2.17	1,255	1,867
18 AWG 7W (0.82 mm²) INDIVIDUAL AND OVERALL SHIELDED PAIRS																				
9250.18021221	18	2	15	0.38	22	12	0.30	40	1.02	0.38	9.7	0.58	14.7	50	1.27	0.69	17.5	0.38	216	322
9250.18041221	18	4	15	0.38	22	12	0.30	50	1.27	0.49	12.4	0.71	18.0	50	1.27	0.82	20.8	0.54	278	414
9250.18061221	18	6	15	0.38	22	12	0.30	50	1.27	0.55	14.0	0.75	19.1	50	1.27	0.86	21.8	0.59	345	513
9250.18081221	18	8	15	0.38	22	12	0.30	50	1.27	0.60	15.2	0.80	20.3	50	1.27	0.92	23.4	0.67	397	590
9250.18101221	18	10	15	0.38	22	12	0.30	60	1.52	0.67	17.0	0.89	22.6	50	1.27	1.00	25.4	0.80	489	727
9250.18121221	18	12	15	0.38	22	12	0.30	60	1.52	0.71	18.0	0.93	23.6	50	1.27	1.04	26.4	0.86	540	803
9250.18161221	18	16	15	0.38	22	12	0.30	60	1.52	0.79	20.1	1.06	26.9	50	1.27	1.17	29.7	1.09	645	959
9250.18201221	18	20	15	0.38	22	12	0.30	60	1.52	0.88	22.4	1.15	29.2	50	1.27	1.26	32.0	1.26	794	1,181
9250.18241221	18	24	15	0.38	22	12	0.30	70	1.78	0.98	24.9	1.24	31.5	50	1.27	1.35	34.3	1.45	907	1,349
9250.18361221	18	36	15	0.38	22	12	0.30	70	1.78	1.15	29.2	1.47	37.3	50	1.27	1.58	40.1	1.99	1,227	1,826
9250.18501221	18	50	15	0.38	22	12	0.30	80	2.03	1.36	34.5	1.69	42.9	60	1.52	1.82	46.2	2.64	1,662	2,473
16 AWG 7W (1.31 mm²) INDIVIDUAL AND OVERALL SHIELDED PAIRS																				
9250.16021221	16	2	15	0.38	22	12	0.30	50	1.27	0.44	11.2	0.67	17.0	50	1.27	0.78	19.8	0.48	260	387
9250.16041221	16	4	15	0.38	22	12	0.30	50	1.27	0.52	13.2	0.71	18.0	50	1.27	0.82	20.8	0.54	334	496
9250.16061221	16	6	15	0.38	22	12	0.30	50	1.27	0.59	15.0	0.84	21.3	50	1.27	0.95	24.1	0.72	443	659
9250.16081221	16	8	15	0.38	22	12	0.30	60	1.52	0.69	17.5	0.93	23.6	50	1.27	1.04	26.4	0.86	515	767
9250.16101221	16	10	15	0.38	22	12	0.30	60	1.52	0.75	19.1	1.02	25.9	50	1.27	1.13	28.7	1.02	616	917
9250.16121221	16	12	15	0.38	22	12	0.30	60	1.52	0.81	20.6	1.06	26.9	50	1.27	1.17	29.7	1.09	684	1,019
9250.16161221	16	16	15	0.38	22	12	0.30	70	1.78	0.95	24.1	1.24	31.5	50	1.27	1.35	34.3	1.45	872	1,298
9250.16201221	16	20	15	0.38	22	12	0.30	70	1.78	1.03	26.2	1.34	34.0	50	1.27	1.45	36.8	1.67	1,024	1,524
9250.16241221	16	24	15	0.38	22	12	0.30	70	1.78	1.10	27.9	1.37	34.8	50	1.27	1.48	37.6	1.74	1,270	1,890
9250.16361221	16	36	15	0.38	22	12	0.30	80	2.03	1.29	32.8	1.60	40.6	60	1.52	1.73	43.9	2.38	1,712	2,547
9250.16501221	16	50	15	0.38	22	12	0.30	80	2.03	1.53	38.9	1.87	47.5	60	1.52	2.00	50.8	3.18	2,215	3,297

PART NUMBER	COND. SIZE (AWG)	NO. OF TRIADS	INSULATION THICKNESS		COMMUNICATION WIRE			INNER JACKET THICKNESS		NOMINAL CORE O.D.		NOMINAL ARMOR O.D.		JACKET THICKNESS		NOMINAL OVERALL O.D.		CROSS-SECTIONAL AREA[1]	APPROXIMATE NET WEIGHT	
					SIZE	INS. THICKNESS														
			mils	mm	AWG	mils	mm	mils	mm	INCHES	mm	INCHES	mm	mils	mm	INCHES	mm	(SQ. IN.)	LBS/1000 FT	kg/1000 m
20 AWG 7W (0.52 mm²) INDIVIDUAL AND OVERALL SHIELDED TRIADS																				
9250.20041222	20	4	15	0.38	22	12	0.30	50	1.27	0.45	11.4	0.67	17.0	50	1.27	0.78	19.8	0.48	263	392
9250.20081222	20	8	15	0.38	22	12	0.30	50	1.27	0.56	14.2	0.80	20.3	50	1.27	0.91	23.1	0.66	376	560
9250.20121222	20	12	15	0.38	22	12	0.30	60	1.52	0.67	17.0	0.89	22.6	50	1.27	1.00	25.4	0.80	514	765
9250.20161222	20	16	15	0.38	22	12	0.30	60	1.52	0.77	19.6	1.02	25.9	50	1.27	1.13	28.7	1.02	616	917
9250.20241222	20	24	15	0.38	22	12	0.30	70	1.78	0.96	24.4	1.24	31.5	50	1.27	1.35	34.3	1.45	869	1,293
9250.20361222	20	36	15	0.38	22	12	0.30	70	1.78	1.11	28.2	1.42	36.1	50	1.27	1.53	38.9	1.86	1,139	1,696
18 AWG 7W (0.82 mm²) INDIVIDUAL AND OVERALL SHIELDED TRIADS																				
9250.18041222	18	4	15	0.38	22	12	0.30	50	1.27	0.54	13.7	0.75	19.1	50	1.27	0.86	21.8	0.59	320	477
9250.18081222	18	8	15	0.38	22	12	0.30	60	1.52	0.69	17.5	0.93	23.6	50	1.27	1.04	26.4	0.86	485	721
9250.18121222	18	12	15	0.38	22	12	0.30	60	1.52	0.79	20.1	1.06	26.9	50	1.27	1.17	29.7	1.09	645	959
9250.18161222	18	16	15	0.38	22	12	0.30	70	1.78	0.90	22.9	1.15	29.2	50	1.27	1.26	32.0	1.26	797	1,185
9250.18241222	18	24	15	0.38	22	12	0.30	70	1.78	1.06	26.9	1.34	34.0	50	1.27	1.45	36.8	1.67	1,119	1,665
9250.18361222	18	36	15	0.38	22	12	0.30	80	2.03	1.29	32.8	1.60	40.6	60	1.52	1.73	43.9	2.38	1,570	2,336
16 AWG 7W (1.31 mm²) INDIVIDUAL AND OVERALL SHIELDED TRIADS																				
9250.16041222	16	4	15	0.38	22	12	0.30	50	1.27	0.58	14.7	0.80	20.3	50	1.27	0.91	23.1	0.58	392	583
9250.16081222	16	8	15	0.38	22	12	0.30	60	1.52	0.79	20.1	1.02	25.9	50	1.27	1.13	28.7	1.02	621	924
9250.16121222	16	12	15	0.38	22	12	0.30	70	1.78	0.95	24.1	1.19	30.2	50	1.27	1.30	33.0	1.34	879	1,308
9250.16161222	16	16	15	0.38	22	12	0.30	70	1.78	1.04	26.4	1.34	34.0	50	1.27	1.45	36.8	1.67	1,074	1,598
9250.16241222	16	24	15	0.38	22	12	0.30	80	2.03	1.27	32.3	1.60	40.6	60	1.52	1.73	43.9	2.38	1,605	2,389
9250.16361222	16	36	15	0.38	22	12	0.30	80	2.03	1.49	37.8	1.83	46.5	60	1.52	1.96	49.8	3.06	2,161	3,216

Dimensions and weights are nominal; subject to industry tolerances.
[1] Cross-sectional area for cable tray fill is in accordance with NEC® Section 392.22.

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